SC2014 : Sensors, Interfacing & Digital Control

Part 2 : Tutorial 0 – Analogue-Digital Filters

Q1 : A cable-based sensing system is used to transmit/receive signal from the remote transducer/sensor. The sensor signal is often corrupted by some high frequency noise. Suppose the signal data has a frequency spectrum of 1 kHz to 2 kHz and the high-frequency noise has a frequency spectrum of 80 kHz to 100 KHz.

Design a suitable *RC* (passive) filter to extract the signal at the receiving end such that the signal gain is maintain at 1 and the noise is attenuated to 0.1 of its initial value. Assume that 0.1 μF capacitors and 0-10 kΩ variable resistors are available for the design.

Low pass filter, signal gain maintained at 1, thus choose 1 decade away for fc == 20kHz
1st order RC filter, fc = 1/(2piRc) RC = 1/40000pi, with C = 0.1uF, R= 1/(40000pi*0.1*10^-6) = 80ohms. Approx attenuation is at -12db, 20 log Attn = -12, thus attn == 0.25.

Q2 : The standard passive filter characteristic is shown below. Identify what is the filter type for this Laplace Equation.

wc = 2pifc = 1/RC, R = 3.2kOhms

sub s = jw, check for response for w =0 and w = infinite, see the ratio approaches 1 or 0. Thus this is High pass filter

$$H(s) = \frac{s}{s + \omega_c}$$

Design the suitable *RC* (passive) filter for a cut-off frequency of 500Hz and its component value. You can assume that 0.1 μF capacitors and 0-10 kΩ variable resistors are available for the design.

Q3 : We can design a Digital Filter based on the Analogue Filter specifications of **Q2**.

continuous time -> s domain, discrete time domain -> z domain

$$H(s) = \frac{s}{s + \omega_c} = \frac{s}{s + 2\pi(500)}$$

1. change to z domain then simplify
2. Ts = 1/5000 = 0.2ms

ans is (10*big term)/(10*big term + pi)

Using bilinear transformation and a sampling frequency of 5000 samples/sec, derive the difference equation (input-output equation).

To get the software implementation, we have to realise that H(z) = Vo(z)/Vi(z). rearrange it to Vo(z) = smt smt*Vi(z)
2. be aware that z^-1 or s^-1 is basically the previous reading.

Q4 : From Tut2 Question2, you have derived the difference equations required for implementing a Low-Pass filter. This is achieved via a Numerical Method known as Backward Euler Method. The solutions include different sampling rates but the same bandwidth (shown below):

Sampling = 1000 Samples/sec, Bandwidth 10Hz

$$V_o[n] = (0.06)V_{in}[n] + (0.94)V_o[n-1]$$

Sampling = 100 Samples/sec, Bandwidth 10Hz

$$V_o[n] = (0.38)V_{in}[n] + (0.62)V_o[n-1]$$

With the Low-Pass 1st Order Low-Pass Digital Filter (via Bilinear Transformation) example. You can also work out the difference equations for the Digital Filter Implementation at the same bandwidth frequency of 10Hz and for two different sampling rate of 1000 sample/sec & 100 sample/sec.

Compare and contrast the difference between the coefficients/terms of the difference equation (input-output equation) and state the pros & cons of the two methods.

Bilinear has a additional term for previous input, while euler has current input and previous output only.

When high sampling frequency, the coeff of previous output has a very high weightage
when smapling frequency is lower, the coeff of previous out dont matter as much
Overall, the bilinear is more accurate, the Euler and Bilinear both heavily relies on previous output. To have higher accuracy for both euler/bilinear, we need to have higher sampling frequency against fc/bandwidth.

Euler is usually for 1st order filters, bilinear is for higher order filter cahracteristics. Euler needs higher sampling frequency at least 20-100 times the signal frequency.